

College of Computer
The Department of Computer Science
CS 451 - 1445H (451)

1091: Introduction to Computer Security (3 units)

Fall 2023

Wednesday 8:00 am to 10:35 am

.....th Week

Lecture # ...



12&13	Network Security : Preparation: Bring an example of at least one network attack. Reading: - Part Five: network and internet security from Cryptography and Network Security - Ch6 from Security in Computing - Provided extra materials Network Security : Preparation:	- 2nd Assignment due 5%
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Network Security

- ◉ Last Lecture :

- ✓ **What is a network?**
- ✓ **Network Topology**
- ✓ **Network Connectivity and Protocols**
- ✓ **Threats to Network Communications**
- ✓ **Interception: Eavesdropping or Wiretapping**
- ✓ **Modification, Fabrication: Data Corruption**

Threats to Network Communications

- **Identity Spoofing**

- ◉ **Identity spoofing attacks allow attackers to impersonate a victim .**

- **Man-in-the-middle attacks.**

- **Message replays**

- **Network spoofing attacks**

- **Software/Service exploitation attacks**

Threats to Network Communications

- Identity Spoofing

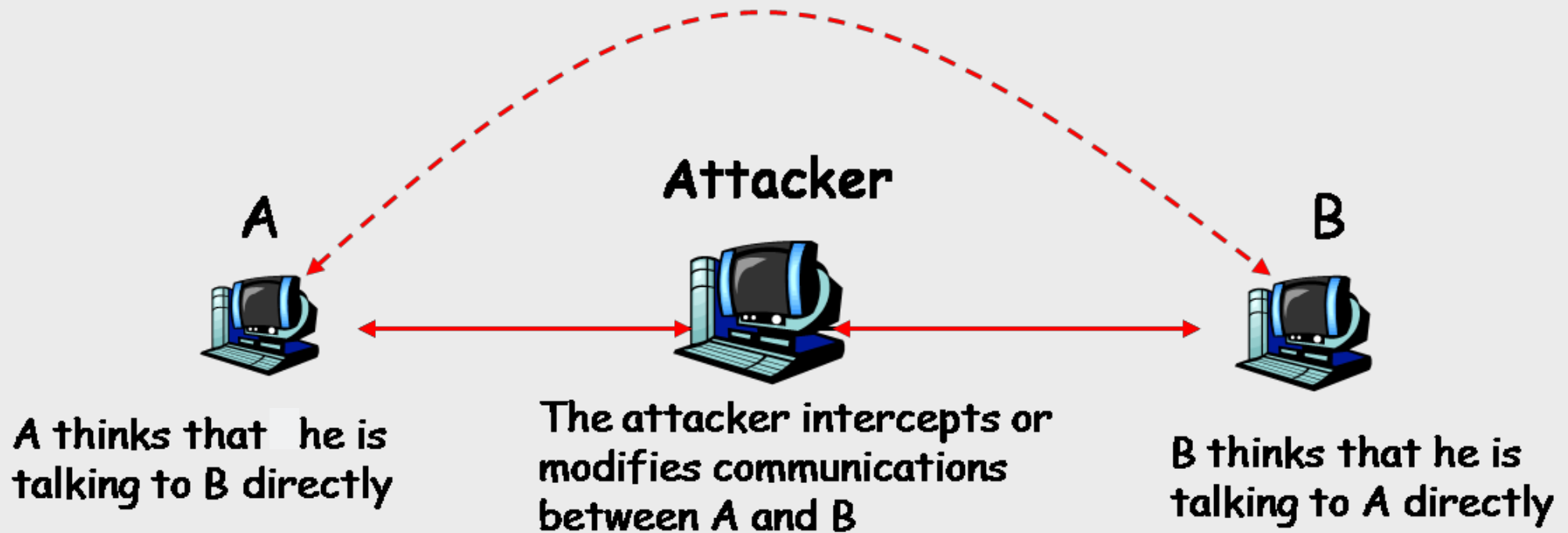
- ◉ **Man-in-the-middle attacks.**

- **Compromise a network device (or installs one of his own) between two or more users. Using this device to intercept, modify, or fabricate data transmitted between users.**

Threats to Network Communications

- Identity Spoofing

- ◉ **Man-in-the-middle attacks.**



Threats to Network Communications

- **Identity Spoofing**

- ◉ **Message Replays**

- **The attacker first intercepts a legitimate message and retransmits it at a later time to the original receiver**

Threats to Network Communications

▸ **Interruption: Loss of Service**

- ◉ **As with the other vulnerabilities we discussed:**
 - **loss of service can be malicious or nonmalicious, intentional or accidental.**
- ◉ **Unlike confidentiality and integrity failures, however, denial of service is not binary <<< ?**
- ◉ **Yes, you do either have service or not, but a critical question is how much**

Threats to Network Communications

▸ **Interruption: Loss of Service**

- ◉ **Excessive Demand :**

- ◉ **Network capacity is enormous but finite**

- ◉ **Thus, with extreme demand an attacker can overwhelm a critical part of a network, from a web page server to a router or a communications line.**

Threats to Network Communications

- **Interruption: Loss of Service**

- ◉ **How the swamped component responds varies:**

- **Some components shut down completely**
- **Some provide degraded (slower) service to all requests,**
- **Others drop some activity in an attempt to preserve service for some users.**

Threats to Network Communications

- **Interruption: Loss of Service**

- ⦿ **Denial-of-service attacks usually try to flood a victim with excessive demand.**
- ⦿ **a successful attacker intent on harming a particular victim often spends time investigating the victim's vulnerabilities and defenses, and plans an appropriate attack.**

Threats to Network Communications

- **Identity Spoofing + Interruption: Loss of Service**

- ◉ **Identity spoofing attacks allow attackers to impersonate a victim .**

- **Network spoofing attacks**

- **Software/Service exploitation attacks**

Threats to Network Communications

- **Identity Spoofing + Interruption: Loss of Service**

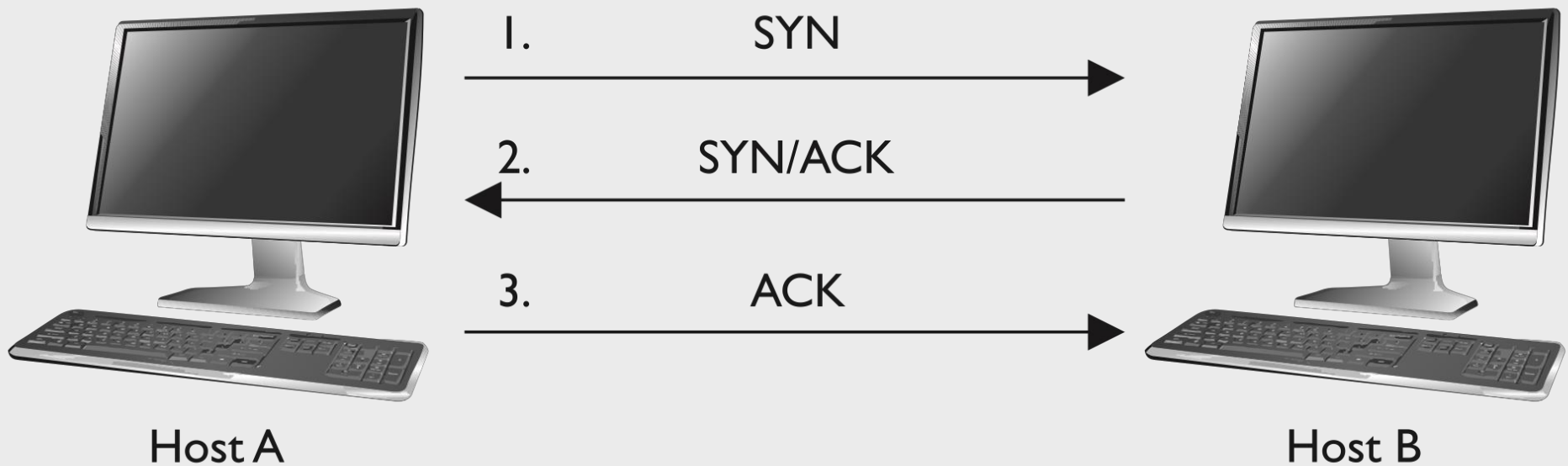
- ◎ **Network spoofing attacks**

- **SYN flooding : The attacker fills the target computer's TCP buffer with a large number of crafted SYN packets**
- **ARP spoofing, which is also known as ARP poisoning**

Network Security

The TCP Handshake

- The TCP Handshake



Threats to Network Communications

▸ Identity Spoofing + Interruption: Loss of Service

◉ SYN flooding

1. **Attacker** sends to **victim** a large number of crafted

SYN packets

2. The **victim's computer** is obliged to send an ACK

packet to the crafted source IP address contained in

the **SYN packet** according to the 3-way handshake

protocol

Threats to Network Communications

▸ Identity Spoofing + Interruption: Loss of Service

◉ SYN flooding

3. Because the source IP address is crafted and unreachable, the **victim computer** will never receive the **ACK packet** it is waiting for, allowing the crafted **SYN packet** to remain in the **TCP buffer**
4. The **TCP buffer** is completely occupied by the crafted SYN packets, causing the computer unable to establish communications with other computers

Threats to Network Communications

▸ Identity Spoofing + Interruption: Loss of Service

◉ TCP Hijacking

V is a company computer

A, an employee of the company, is going to remote logon to **V**

1. **A** sends a SYN packet to **V**
2. The **attacker** intercepts this packet, and uses SYN flooding to mute **V** so that **V** can't complete the three-way handshake with **A** (his computer)
3. The **attacker** predicts the correct TCP sequence number for the ACK supposed to be sent from **V** to **A**. The **attacker** then crafts an ACK packet with the sequence number and attacker's IP address and sends it to **A**

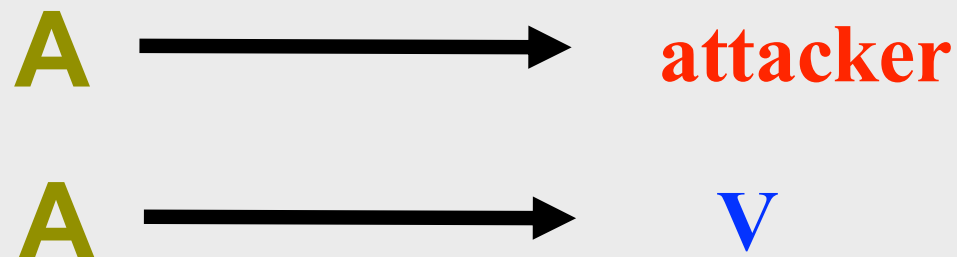
Threats to Network Communications

▸ Identity Spoofing + Interruption: Loss of Service

◉ TCP Hijacking

4. **A** verifies the ACK packet and sends an ACK packet to the **attacker** to complete this handshake

5. The TCP connection is established between **A** and the **attacker**, instead of between **A** and **V**.



Threats to Network Communications

- **Identity Spoofing + Interruption: Loss of Service**

- ◉ **ARP Spoofing**

- **The attacker changes the legitimate MAC**

address of a networked computer to a different

MAC address chosen by the attacker

Network Security

▸ Denial of Service (DoS)

- ◉ A denial-of-service, or DoS, attack is an attempt to defeat availability
- ◉ **Denial of service** means just what its name implies: **a user is denied access to authorized services or data.**
- ◉ Confidentiality and integrity are concerned with **preventing unauthorized access;**
- ◉ Availability is concerned with **preserving authorized access.**

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- **Denial of Service (DoS)**

- ⦿ **Confidentiality and integrity tend to be binary:**

Data or objects either are or are not kept private and unmodified;

- ⦿ **Availability can be more nuanced, in that there may be service but in insufficient quantity or at unacceptable responsiveness.**

Network Security

- **Denial of Service (DoS)**
- ◉ **What do you think that might deny access in a computer network ?**
- ◉ **One potential weakness is the capacity of the system**
 - **also known as volume-based or volumetric attacks**
- ◉ **an attack can exhaust the application that services a particular network**
 - **is called an application- based attack.**

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▸ Denial of Service (DoS)

- ◉ cut or disable the communications link between two points
- ◉ a hardware or software failure
- ◉ DOS can occur from excessive volume, a failed application, a severed link, or hardware or software failure.

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- **Insufficient capacity**

- ◉ **Example about insufficient capacity :**

- **Flooding**
- **Blocked Access**
- **Access Failure**

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▸ Flooding Attacks in Detail

- ◉ Flooding occurs because the incoming bandwidth is insufficient or resources—hardware devices, computing power, software, or table capacity—are inadequate

- Insufficient Resources

- Insufficient Capacity

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▸ Flooding Attacks in Detail

- 1. Network Flooding Caused by Malicious Code**
- 2. Network Flooding by Resource Exhaustion**
- 3. Denial of Service by Addressing Failures**
- 4. Traffic Redirection**
- 5. DNS Attacks**
- 6. Exploiting Known Vulnerabilities**
- 7. Physical Disconnection**

Network Security

▸ Flooding Attacks in Detail

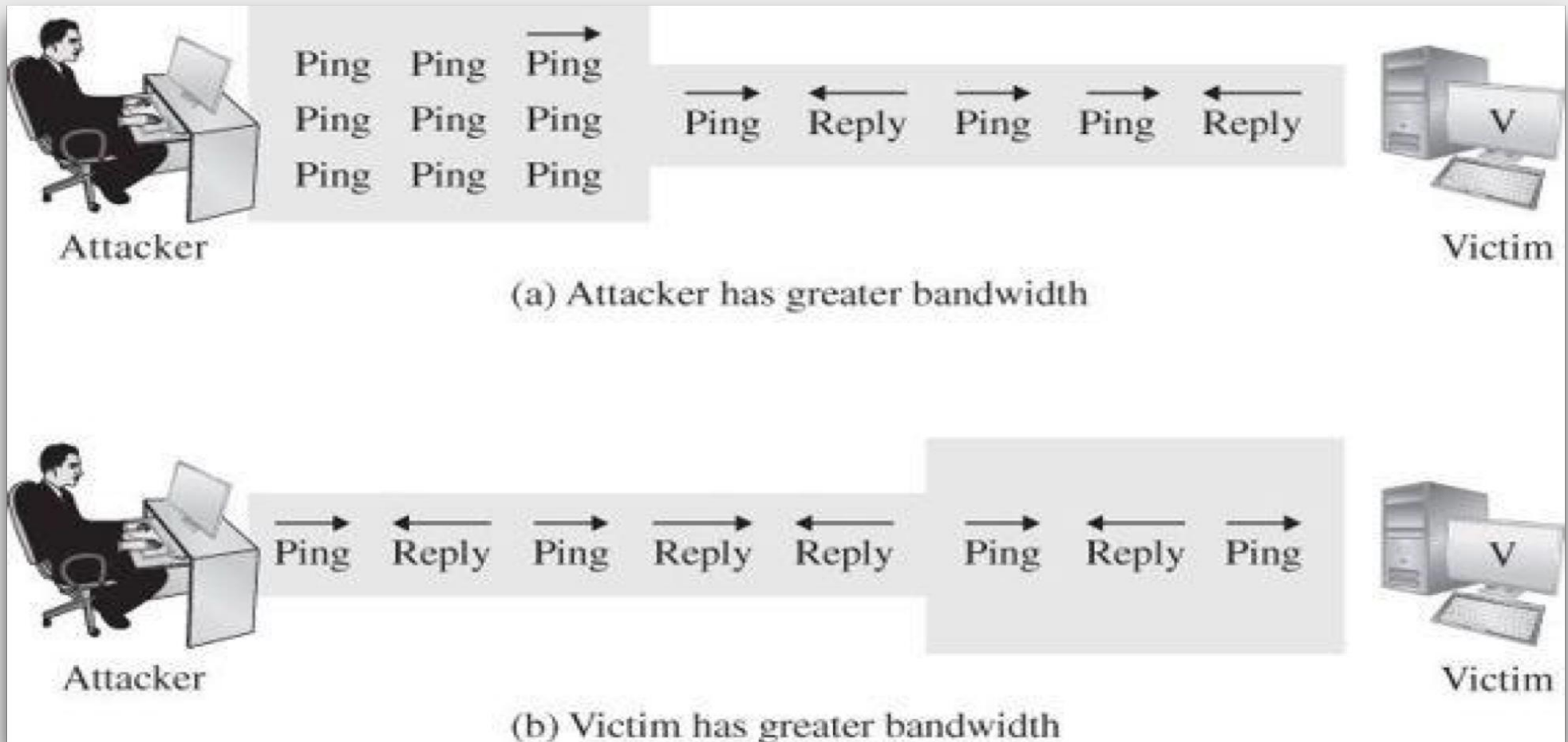
1. Network Flooding Caused by Malicious Code

- ⦿ The most primitive denial-of-service attack is flooding a connection
- ⦿ TCP and UDP, there is a third class of protocols, called ICMP or Internet Control Message Protocols.
 - ⦿ ICMP protocols include:
 - ping
 - echo
 - source quench

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▸ Flooding Attacks in Detail

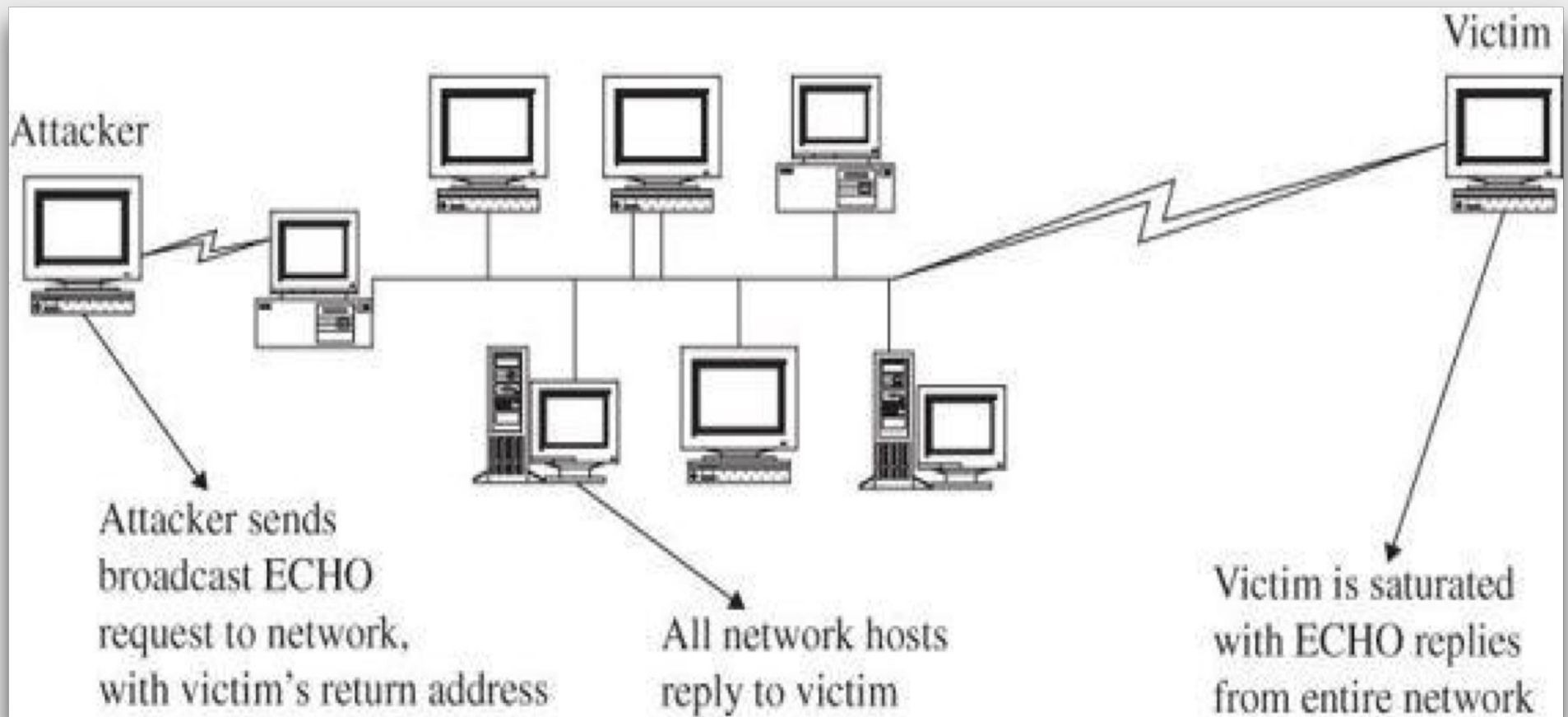
◉ Ping of Death



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▸ Flooding Attacks in Detail

◉ Smurf



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▸ Flooding Attacks in Detail

- ◉ Echo–Chargen
- ◉ SYN Flood

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▸ Flooding Attacks in Detail

2. Network Flooding by Resource Exhaustion

- context switching
- thrashing
- ⦿ IP Fragmentation: Teardrop

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- **Flooding Attacks in Detail**

3. Denial of Service by Addressing Failures

- **DNS Poisoning**
- **DNS Spoofing**

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▸ Flooding Attacks in Detail

3. Denial of Service by Addressing Failures

▸ DNS poisoning

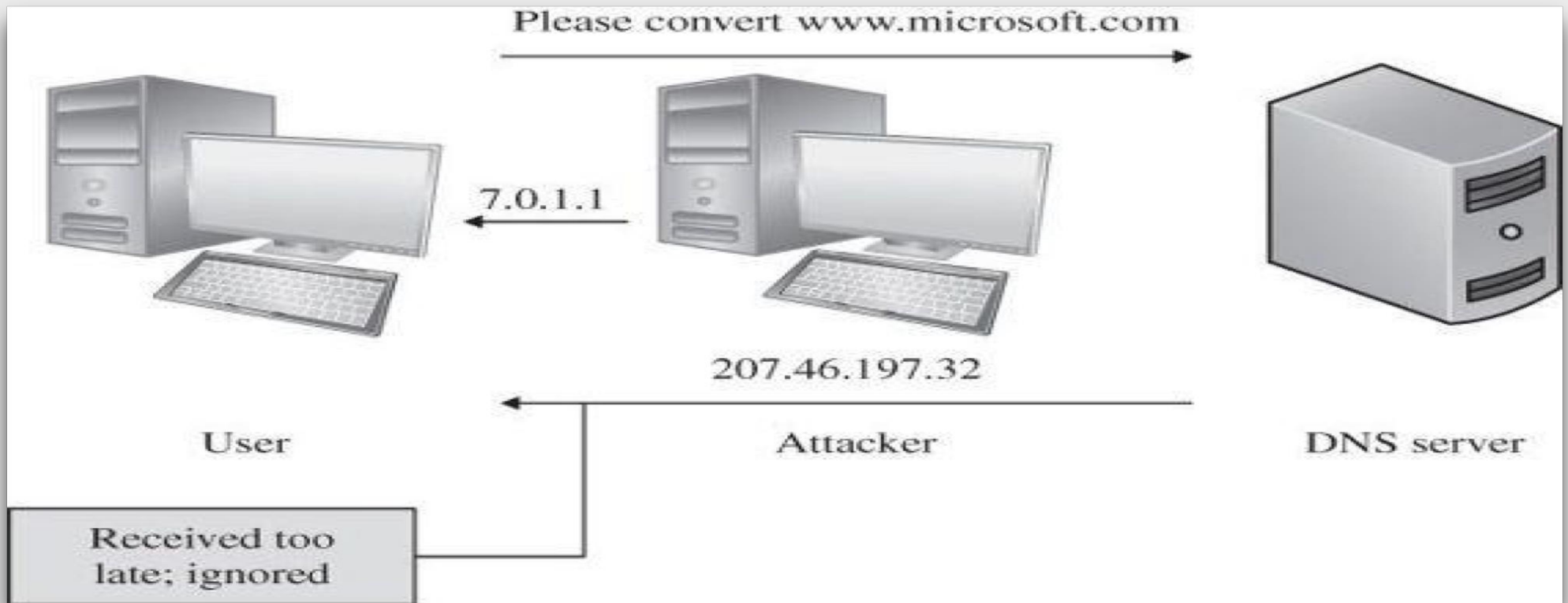


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▸ Flooding Attacks in Detail

3. Denial of Service by Addressing Failures

▸ DNS Spoofing



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▸ Flooding Attacks in Detail

3. Denial of Service by Addressing Failures

- DNS Spoofing
- Any server can respond to a DNS lookup request; the first responder wins. Being first lets an attacker redirect traffic.
- Rerouting Routing
- Router Takes Over a Network

Network Security

▸ Flooding Attacks in Detail

3. Denial of Service by Addressing Failures

▸ Source Routing and Address Spoofing

4. Traffic Redirection

5. DNS Attacks

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▸ Flooding Attacks in Detail

5. DNS Attacks

- Session Hijack
- DNS Cache Poisoning

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▸ Flooding Attacks in Detail

6. Exploiting Known Vulnerabilities

- A zero-day exploit

7. Physical Disconnection

- Transmission Failure
- Component Failure

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▸ Distributed Denial-of-Service

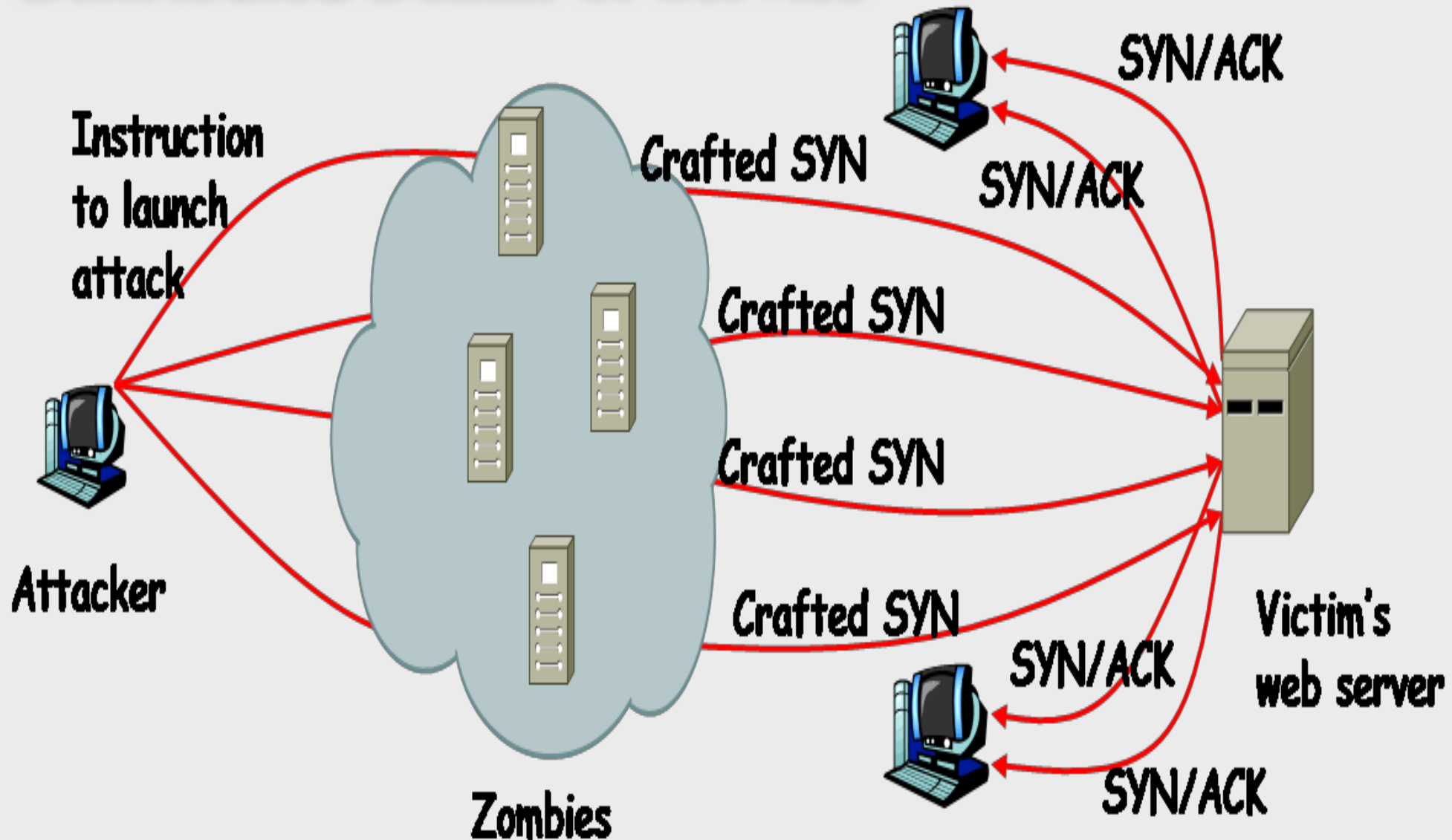
◉ To mount a distributed denial-of-service (or DDoS)

attack, an attacker does two things :

1. The attacker wants to conscript an army of compromised machines to attack a victim “**zombie**”
2. The attacker chooses a victim and sends a signal to all the zombies to launch the attack.

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▸ Distributed Denial-of-Service



Network Security

▸ **Scripted Denial-of-Service Attacks**

- ◉ **Easily launched from scripts**
- ◉ **DDoS attack tools first appeared in mid-1999**
- ◉ **DDoS tools include Tribal Flood Network (TFN), Trin00, and TFN2K**
- ◉ **As new vulnerabilities that allow Trojan horses to be planted are discovered and as new denial-of-service attacks are found**

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- **Scripted Denial-of-Service Attacks**

- ⦿ **According to the U.S. Computer Emergency**

Response Team (CERT)

- ⦿ **Windows-based machines are becoming more popular targets for attack than other systems.**

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▸ **Scripted Denial-of-Service Attacks**

◉ **Bots :**

- **Zombies (or bots, hackers for robots) are machines running pieces of malicious code under remote control. These code objects are Trojan horses that are distributed to large numbers of victims' machines. Because they may not interfere with or harm a user's computer (other than consuming computing and network resources), they are often undetected.**

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- **Scripted Denial-of-Service Attacks**

- ◉ **Botnets :**

- **Networks of bots, are used for massive denial-of-service attacks, implemented from many sites working in parallel against a victim.**
- **They are also used for spam and other bulk email attacks**

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- **Scripted Denial-of-Service Attacks**
- ◉ **Botnet Command and Control Update**
 - **A network of bots requires a command hierarchy; the bots require officers to tell them when to attack, against whom, and with what weapon**
 - **The bot headquarters is called a command-and-control center.**

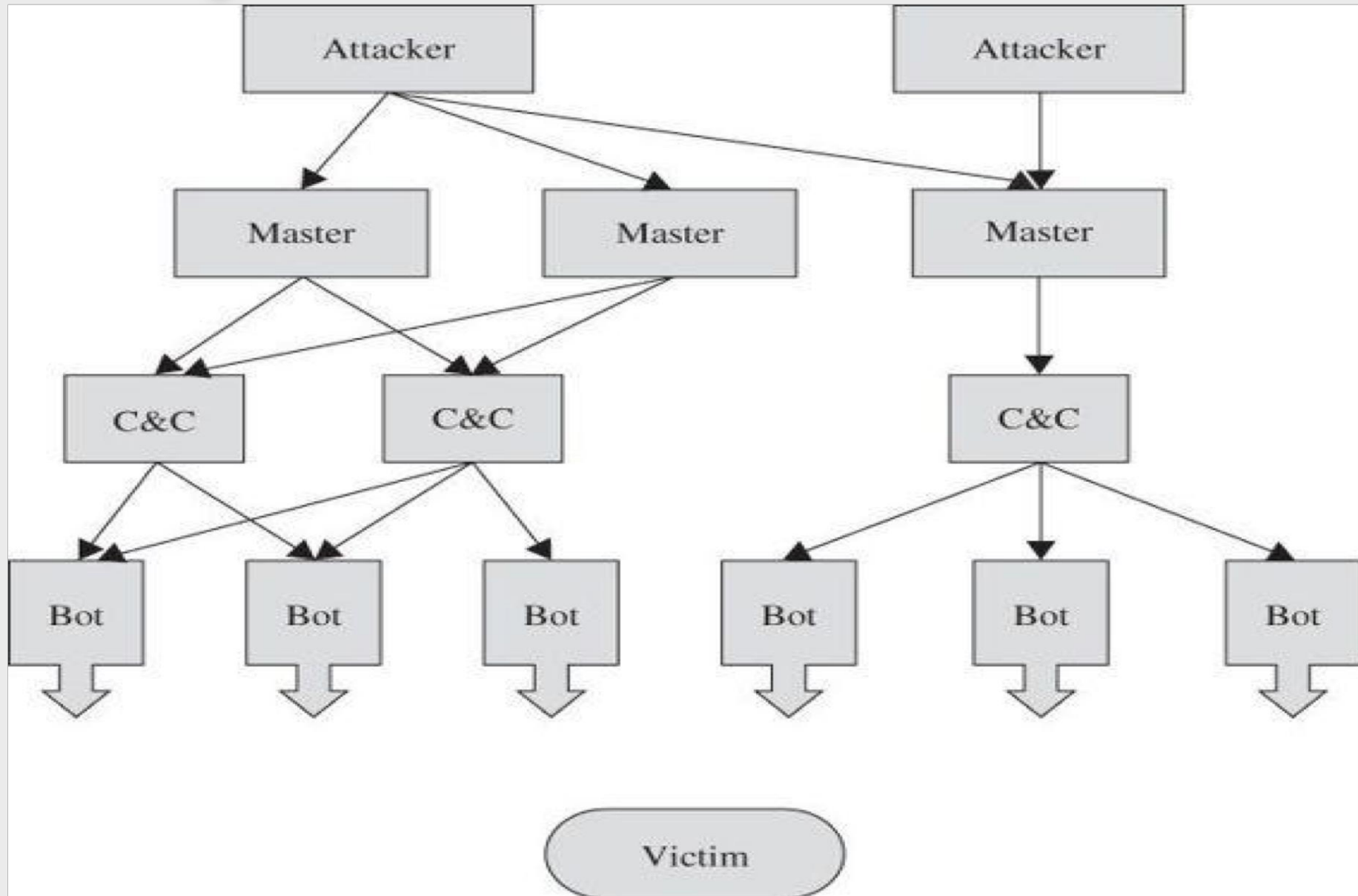
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- **Scripted Denial-of-Service Attacks**

- ⦿ **A botnet command-and-control center instructs specific machines to target a particular victim at a given time and duration.**

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Scripted Denial-of-Service Attacks



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▸ Scripted Denial-of-Service Attacks

◉ Rent-A-Bot

▸ People who infect machines to turn them into bots are called botmasters

◉ A botmaster often has two uses for the botnet:

1. the botnet should be available for attacks when the botmaster wants to go after a victim

2. botmasters also sometimes rent out their botnets to others.

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- **Scripted Denial-of-Service Attacks**

- ◉ **Opt-In Botnets**

- **Join with a group of like-minded individuals to launch a distributed denial-of-service attack against the outrage.**

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▸ **Scripted Denial-of-Service Attacks**

◉ **Malicious Autonomous Mobile Agents**

- **Bots belong to a class of code known more generally as malicious autonomous mobile agents.**
- **Working largely on their own, these programs can infect computers anywhere they can access, causing denial of service as well as other kinds of harm.**
- **an inoculation agent : to set up the process and, usually, to establish a scheme for updates.**

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▸ Scripted Denial-of-Service Attacks

◉ Coping with DDoS Attacks

- adjusting the number of active servers
- load balancing (evening the computing load across available servers)
- shunning (reducing service given to traffic from certain address ranges)
- blacklisting (rejecting connections from certain addresses)

Wireless Network Security

▸ Vulnerabilities in Wireless Networks

- ◉ Wireless communication will never be as secure as wired, because the exposed signal is more vulnerable.
- ◉ Wireless networks are subject to threats to **confidentiality**, **integrity**, and **availability**
- ◉ The attacker can either join the network of the target and participate in data exchanges, or merely observe the traffic as a bystander.

Wireless Network Security

- **Vulnerabilities in Wireless Networks**

- ◉ **Confidentiality**

- ◉ **Certainly, if data signals are transmitted in the open, unintended recipients may be able to get the data.**

Wireless Network Security

- **Vulnerabilities in Wireless Networks**

- ◉ **Integrity**

- ◉ **As for integrity, we must consider both malicious and nonmalicious sources of problems.**

- **Nonmalicious sources**

- **Malicious attacks**

Wireless Network Security

▸ Vulnerabilities in Wireless Networks

◉ Availability

◉ Availability involves three potential problems :

- First, the most obvious, occurs when a component of a wireless communication stops working because hardware fails, power is lost, or some other catastrophe strikes
- A second problem of availability is loss of some but not all access
- a third problem: the possibility of rogue network connection.

Wireless Network Security

▸ Specific wireless attacks and countermeasures.

◉ Unauthorized WiFi Access

- ◉ An unauthorized device can attempt to establish an association with an access point.
- ◉ Remember from the WiFi protocols that access basically involves three steps:
 1. The access point broadcasts its availability by sending a beacon, an invitation for devices to connect with it.
 2. A device's NIC responds with a request to authenticate, which the access point accepts.
 3. The device's NIC requests establishment of an association, which the access point negotiates and accepts.

Wireless Network Security

- **Specific wireless attacks and countermeasures.**
- ◉ **There are threats at each of these points.**
 - 1. Anyone can pick up and reply to a broadcast beacon**
 - 2. The authentication is not rigorous; in basic WiFi mode the access point accepts any device, without authentication**
 - 3. Any access point can accept an association with any device**

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Thank you

Q & A